

1 The cells in a tissue are often at different stages of the cell cycle.

(a) Fig. 1.1 shows cells at different stages of the cell cycle.

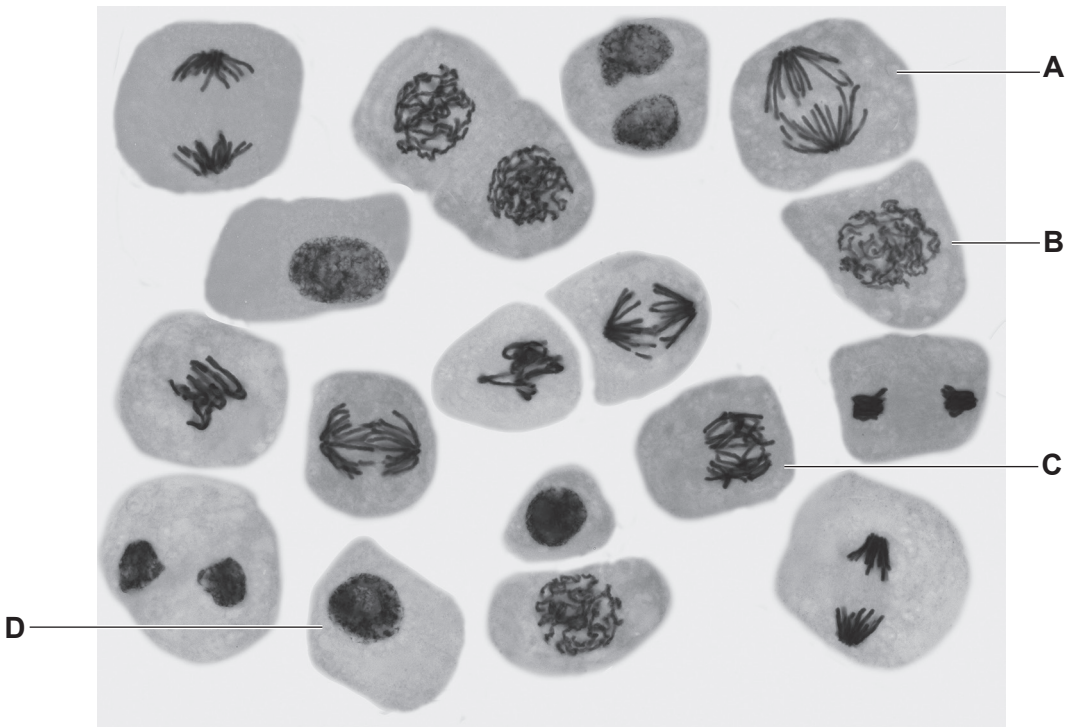


Fig. 1.1

(i) Identify the stages of mitosis occurring in the cells labelled **B** and **C** in Fig. 1.1.

B

C [2]

(ii) Describe the behaviour of the chromosomes in the stage of mitosis shown in cell **A** in Fig. 1.1.

.....
.....
.....
.....
.....
.....
..... [3]

- (iii) Cell **D** in Fig. 1.1 is in interphase.

Describe the role of DNA ligase in interphase.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) Telomerase is an enzyme that is active during interphase in some cells.

Telomerase helps to maintain telomeres present on chromosomes.

- (i) State the location of telomeres on a chromosome.

.....

..... [1]

- (ii) Some people with a long lifespan have cells showing a higher than normal activity of telomerase.

Suggest why a long lifespan could result from a higher telomerase activity.

.....

.....

.....

.....

..... [2]

[Total: 11]